

Henry Liu

Curriculum Vitae

(Last updated: December 15, 2022)

Mathematical Institute
University of Oxford
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Research interests

Algebraic geometry, geometric representation theory, and mathematical physics. Specific focuses: enumerative geometry, moduli of sheaves and quiver representations, and their (quantum) symmetry algebras e.g. vertex algebras and quantum groups.

Positions

Postdoctoral Research Associate in Geometry, University of Oxford 2021 – 2023
Member of the Simons Collaboration on Special Holonomy in Geometry, Analysis, and Physics

Education

Ph.D., Mathematics, Columbia University 2016 – 2021

Advisor: Andrei Okounkov

Thesis: Asymptotic representations of shifted quantum affine algebras from critical K-theory.

Bachelor of Mathematics, University of Waterloo 2012 – 2016
Honours Pure Mathematics & Computer Science

Publications, preprints, and code

- [1] Henry Liu. Multiplicative vertex algebras and quantum loop algebras. October 2022. [arXiv:2210.04773](#), submitted.
- [2] Henry Liu. Equivariant K-theoretic enumerative invariants and wall-crossing formulae in abelian categories. July 2022. [arXiv:2207.13546](#), submitted.
- [3] Henry Liu. A representation-theoretic approach to qq -characters. *SIGMA, Symmetry Integrability Geom. Methods Appl.*, 18(paper 090, 20), November 2022. [arXiv:2203.07072](#).
- [4] Henry Liu. Box-counting code. <https://github.com/liu-henry-hl/boxcounting>, 2022.
- [5] Henry Liu. Quasimaps and stable pairs. *Forum of Mathematics, Sigma*, 9(e32), 2021. [arXiv:2006.14695](#).

- [6] Henry Liu. *Asymptotic representations of shifted quantum affine algebras from critical K-theory*. PhD thesis, Columbia University, 2021. Available from [Academic Commons](#).
- [7] Henry Liu. Self-duality in quantum K-theory. June 2019. [arXiv:1906.10824](#), preprint.
- [8] Gabriel Coutinho and Henry Liu. No Laplacian perfect state transfer in trees. *SIAM J. Discrete Math.*, 29(4):2179–2188, 2015. [arXiv:1408.2935](#).
- [9] Henry Liu, Mingbin Xu, Ziting Yu, Vincent Corvinelli, and Calisto Zuzarte. Cardinality estimation using neural networks. In *Proceedings of the 25th Annual International Conference on Computer Science and Software Engineering*, CASCON '15, pages 53–59, USA, 2015. IBM Corp.

Honors and awards

Columbia Dean's Fellow	2016 – 2021
NSERC Undergraduate Student Research Award	2015
IBM CASCON 2015 Best Student Paper	2015
NSERC Undergraduate Student Research Award	2014
University of Waterloo Mathematics National Scholarship	2012 – 2016
University of Waterloo Dean's Honour List	2012 – 2016
University of Waterloo President's Scholarship	2012

Invited talks

Algebra and number theory seminar, University of Glasgow	Nov 2022
EGRET seminar, University of Edinburgh	Nov 2022
Algebra seminar, University of Oslo	Nov 2022
MAGIC seminar, Imperial College London	Oct 2022
Informal String-Math Seminar, UC Berkeley	Sep 2022
<i>Multiplicative vertex algebras and wall-crossing in equivariant K-theory</i>	
Progress and Open Problems 2022, Simons Collaboration on Special Holonomy, Stony Brook	Sep 2022
<i>Multiplicative vertex algebras and wall-crossing in equivariant K-theory</i>	
Workshop: Integrability, Enumerative Geometry and Quantization	Aug – Sep 2022
Simons Center for Geometry and Physics, Stony Brook	
<i>2 x 1.5 hour problem sessions</i>	Aug 2022
<i>Multiplicative vertex algebras and wall-crossing in equivariant K-theory</i>	Sep 2022
Enumerative geometry seminar, Tsinghua Yau Mathematical Sciences Center	Mar 2022
<i>Quasimaps and stable pairs</i>	
Geometry and Analysis seminar, Oxford	Nov 2021
<i>Quasimaps and Donaldson–Thomas theory</i>	
Algebraic Geometry seminar, University of British Columbia	Jan 2021
<i>Quasimaps and stable pairs</i>	

Joint Differential Geometry seminar, CUHK-Harvard-YMSC <i>Self-duality in quantum K-theory</i>	Oct 2020
Informal mathematical physics seminar, Columbia University <i>Computational aspects of Donaldson–Thomas theory</i>	Oct 2020
Mathematical physics seminar, Perimeter Institute <i>Quasimaps and BPS counts</i>	Oct 2020
Informal mathematical physics seminar, Columbia University <i>Quasimaps and stable pairs</i>	June 2020
Combinatorics & Optimization undergraduate research seminar, University of Waterloo <i>Perfect state transfer on graphs</i>	Jul 2014

Service and organization

Referee: Geometry & Topology

Reviewer: zbMATH

Mentor: Columbia Directed Reading Program 2020

(Co-)Organization:

<i>Quivers, Clusters, Moduli and Stability</i> conference	Jan 2023
Oxford <i>Geometry and Analysis</i> seminar	2021 – Present
Columbia student learning seminars on: category $\mathcal{O}$, conformal field theory, symplectic duality, GW/DT correspondence, enumerative geometry	2017 – 2021
Waterloo student seminar on QFT, string theory, and mirror symmetry	2015 – 2016

Teaching

Instructor (University of Oxford):

<i>Equivariant K-theory</i> (Taught Course Center, 16 x 1-hour lectures)	Apr – Jun 2022
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Instructor (Columbia University):

<i>Ordinary Differential Equations</i> (UN3027)	Summer 2019
<i>Calculus I</i> (UN1101)	Spring 2019
<i>Geometry and Topology</i> , Science Honors Program	Fall 2019 – Spring 2020
<i>Representation Theory and Physics</i> , Science Honors Program	Fall 2016 – Spring 2019

Section leader (Columbia University):

<i>Undergraduate Seminar on Knot Theory</i> (UN3952)	Spring 2020
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Teaching assistant (Columbia University): 2016–2021

*Making, Breaking Codes, Ordinary Differential Equations, Linear Algebra,
Intro to Modern Algebra II, Intro to Modern Algebra I.*

Teaching assistant (University of Waterloo): 2013–2016

Linear Algebra I, Calculus II, Calculus III, Intro to Combinatorics

Industry experience

LinkedIn Software Engineering Intern

Summer 2015

IBM EPIC Software Engineering Intern

Summer 2013

Patents

Vincent Corvinelli, Henry Liu, Mingbin Xu, Ziting Yu, and Calisto Zuzarte. *Selectivity estimation using artificial neural networks*. United States Patent US 10,318,866. June 11, 2019.

Languages

Native: English.

Fluent (non-technical): Mandarin Chinese.

Elementary: French, German.

Technology (> 5 years experience): C/C++, Python, SageMath, L^AT_EX.